

Question	Answer	Mark	Mark scheme	Additional guidance
23	20	B1	stating bound, eg 10.65 or 10.55 or 31 min 48.5sec or 31 min 47.5sec or 1908.5sec or 1907.5sec	
		P1	(dep on B1) for a correct bound for time in hours, eg 0.5301(38...) or 0.5298(61...) Or a correct process to find one bound for speed in km per minute eg [UB of S] = [UB of D]÷[LB of T] or [LB of S] = [LB of D]÷[UB of T] or a correct process to find one bound for speed in km per second eg [UB of S] = [UB of D]÷[LB of T] or [LB of S] = [LB of D]÷[UB of T]	Bound rounded or truncated to 4 dp Where $10.6 < [\text{UB of D}] \leq 10.65$ and $31 \text{ min } 47.5 \text{ sec} \leq [\text{LB of T}] < 31 \text{ min } 48 \text{ sec}$ Or $10.55 \leq [\text{LB of D}] < 10.6$ and $31 \text{ min } 48 \text{ sec} < [\text{UB of T}] \leq 31 \text{ min } 48.5 \text{ sec}$ Where $10.6 < [\text{UB of D}] \leq 10.65$ and $1907.5 \text{ sec} \leq [\text{LB of T}] < 1908 \text{ sec}$ Or $10.55 \leq [\text{LB of D}] < 10.6$ and $1908 \text{ sec} < [\text{UB of T}] \leq 1908.5 \text{ sec}$
		P1	(dep on P1) for correct process to find one bound for speed in km per hour, eg [UB of S] = [UB of D]÷ 0.5298(61...) or [LB of S] = [LB of D]÷ 0.5301(38...) OR Correct process to convert a bound for speed in km per minute to km per hour eg [UB of S] = [UB of D]÷[LB of T] × 60 or [LB of S] = [LB of D]÷[UB of T] × 60 OR Correct process to convert a bound for speed in km per second to km per hour eg [UB of S] = [UB of D]÷[LB of T] × 60 × 60 or [LB of S] = [LB of D]÷[UB of T] × 60 × 60	Time used in hours Where $10.6 < [\text{UB of D}] \leq 10.65$ and $31 \text{ min } 47.5 \text{ sec} \leq [\text{LB of T}] < 31 \text{ min } 48 \text{ sec}$ Or $10.55 \leq [\text{LB of D}] < 10.6$ and $31 \text{ min } 48 \text{ sec} < [\text{UB of T}] \leq 31 \text{ min } 48.5 \text{ sec}$ Where $10.6 < [\text{UB of D}] \leq 10.65$ and $1907.5 \text{ sec} \leq [\text{LB of T}] < 1908 \text{ sec}$ Or $10.55 \leq [\text{LB of D}] < 10.6$ and $1908 \text{ sec} < [\text{UB of T}] \leq 1908.5 \text{ sec}$
		A1	for both correct bounds from correct working, 20.099... and 19.900...	Figures rounded or truncated to 3 sf or better
		C1	for 20 correct to 2 significant figures as both bounds agree.	